Matrix: a closed-loop adaptive-radiotherapy harness on a synthetic digital twin, grounded on real anatomy and dose geometry

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Abstract. Adaptive radiotherapy is moving toward closed control loops that re-plan dose from imaging acquired during treatment. Before such a loop can be trusted, two things must be demonstrated separately: that the *control mechanics* close, and that a *trust gate* can withhold action where the imaging-derived estimate is unreliable. We build **Matrix**, a four-stage closed loop — scan → posterior → trust gate + action gate → dose re-plan → re-scan — on a *synthetic digital twin* with known ground-truth intravoxel incoherent motion (IVIM) parameters (D, D^*, f) and a dose/response model. The loop closes reproducibly: across 6 iterations on a 12×12 twin, a trust gate keyed on the calibrated perfusion-fraction error bar holds the *action* rate on untrustworthy (low-SNR) voxels at 0.000 [P] (vs 0.422 [0.362, 0.486] [P] ungated) while trustworthy voxels stay free, and trusted-tumour perfusion falls 0.176 [0.167, 0.185] [P] (95% bootstrap CI excludes 0) as the treatment decision winds down (21 → 3 → 0 → 0 → 0 → 0). The gate's separation is perfect by construction (AUROC 1.00): the twin builds the untrustworthy zone, so this is a property of the synthetic design, not a performance claim. We then *ground* the loop on **real** anatomy and a **real** delivered-dose grid from a public radiotherapy dataset (TCIA Pancreatic-CT-CBCT-SEG, CC BY 4.0) through an interface-only substrate swap that leaves the loop engine byte-for-byte unchanged. The grounded loop closes (trusted-tumour drop 0.169 [0.166, 0.171] [P]), and it surfaces the paper's central, honest negative: on real *delivered* dose, "holding" an untrusted voxel no longer protects it — its perfusion still drops 0.148 [0.139, 0.156] (vs 0.000 on the synthetic baseline), because the gate suppresses only *new* actions, not dose already delivered. Action-suppression is not outcome-protection. This is a methods demonstration: it validates the harness and a trust-gating discipline, not any clinical effect.

Scope (read first). Matrix is a *working closed-loop harness on a synthetic digital twin*, **not** a validated clinical loop. Every result means "the loop closes and behaves sensibly" — on a synthetic twin, or (with the **Ferry** substrate) on *real anatomy and real dose geometry with synthetic perfusion*. There is no scanner and no real diffusion data; **no clinical claim is made anywhere**. Calibration, trust, and convergence numbers consume **Fashion** (the calibration ruler) and **Minos** (the trust/action gates), both in review; every such number is marked [P] and is regenerated from seeded results by `consistency.py`. **Forge** (the dose engine) is not built (deferred 2027) and is presented as drop-in future work, not a dependency met.

1. Introduction

Magnetic-resonance-guided adaptive radiotherapy (MRgRT) raises a control-systems question that precedes any clinical one: can a re-planning loop be made to act *only where the evidence supports acting*? A loop that re-plans dose from quantitative imaging acquired mid-treatment inherits all of that imaging's uncertainty. Where a parameter is well identified the loop should be free to act; where it is not, the loop should withhold action rather than chase noise. The pre-requisite for studying that discipline is a loop whose mechanics are known to close and a trust gate whose firing can be checked against ground truth — neither of which requires, or should be confused with, a real clinical result.

This paper builds and validates exactly that pre-requisite. **Matrix** is a closed-loop harness on a *synthetic digital twin*: a simulated abdominal target with known ground-truth IVIM parameters (D, D^*, f) — where f, the perfusion fraction, is the quantity of interest (QoI) — and a dose/response model. The loop runs four stages every iteration,

$$\text{scan} \;\rightarrow\; \text{posterior} \;\rightarrow\; \text{trust gate} + \text{action gate} \;\rightarrow\; \text{dose re-plan} \;\rightarrow\; \text{(re-scan)},$$

and the twin evolves under the delivered plan so that the next scan sees the consequence of the last decision. Because the twin's ground truth is known and seeded, every behaviour of the loop is falsifiable: we can ask whether the gate fires where the twin made the estimate untrustworthy, and whether the loop converges, and answer with confidence intervals against truth.

Matrix is deliberately scoped as a *no-scanner* run-mode of a larger capstone ("Keystone") that runs the same loop in real time. Building Matrix first de-risks the control logic independently of scanner, dose-engine, and regulatory access: the capstone exists as a working artifact even if those never land. The cost of that choice is that *nothing here is a clinical claim* — a discipline we enforce structurally (Sec. 7) rather than rhetorically.

We make three contributions. (i) A reproducible four-stage closed loop on a synthetic twin, with a trust gate that withholds action where the calibrated error bar is untrustworthy, validated end-to-end with bootstrap confidence intervals (Sec. 4). (ii) An *identifiability* result: under the textbook segmented IVIM fit, the pseudo-diffusion coefficient D^* is not recoverable, so the physically-motivated "high-D^*" sub-region cannot drive the trust gate; the gate is instead keyed on the *measured* inflation of the perfusion-fraction error bar σ_f (Sec. 4.6). (iii) A grounding of the loop on *real* anatomy and *real* dose geometry through an interface-only substrate swap (**Ferry**) that leaves the loop engine byte-for-byte unchanged, which surfaces the paper's central honest negative: *action-suppression is not outcome-protection* on a real prescription (Sec. 5).

2. Background and positioning

IVIM perfusion as the QoI. The intravoxel-incoherent-motion model decomposes the diffusion-weighted signal into a tissue-diffusion term and a fast "pseudo-diffusion" perfusion term, parameterised by (D, D^*, f). The perfusion fraction f is a candidate biomarker of tumour vascularity and is the QoI the loop acts on. It is also notoriously ill-conditioned: D^* and f are coupled, and a segmented two-step fit — the standard estimator — recovers D^* poorly. Matrix treats that ill-conditioning as a feature to be *detected*, not hidden (Sec. 4.6).

Three consumed components, each behind a clean interface. The loop consumes three components, none of which is final, so each sits behind a documented interface with a clearly-labelled placeholder; the real component drops in *without touching the loop engine*.

- **Fashion** — the calibration ruler that turns a raw posterior spread into a *calibrated* error bar with a coverage/ECE read-out (interface `Ruler.calibrate`; placeholder `PlaceholderRuler`, NOT-Fashion). In review at *NMR in Biomedicine*; no DOI assigned.
- **Minos** — the trust gate (value of a trusted gate, VoTG) and the treat/spare/escalate action gate (interfaces `TrustGate`, `ActionGate`; placeholders NOT-Minos). Theory half machine-verified; applied half in review.
- **Forge** — the Monte-Carlo dose engine (interface `DoseEngine`; placeholder `PlaceholderDoseEngine`, NOT-Forge). **Not built** — Forge today is only a dose-simulation feasibility benchmark; the re-plan engine is deferred to 2027. Forge is therefore presented as drop-in future work, not a dependency met.

Every number that rides on Fashion or Minos is marked [P]. Forge's placeholder affects only the *physical realism* of the re-plan, not whether the loop closes or how the gates behave; it is not a release condition (Sec. 7).

The synthetic/real boundary. Matrix's default substrate is fully synthetic. The optional **Ferry** substrate (Sec. 5) makes *anatomy* and *dose geometry* real (a public RT dataset), while *perfusion/IVIM stays synthetic* — there is no scanner, hence no real diffusion data. We state which loop quantities are real and which are synthetic at every results claim, and never blur the two.

3. Methods

3.1. The synthetic digital twin

The twin is a 12×12 abdominal slice (144 voxels) with a central tumour disk, an organ-at-risk corner, and normal tissue elsewhere. Per-voxel ground-truth (D, D^*, f) are drawn from tissue priors with seeded jitter; a reproducible subset of tumour voxels is pushed into a high-D^* regime (a vigorous-perfusion core). All randomness derives from a single master seed (20260622), so two twins built from the same seed are bit-for-bit identical in every field — the reproducibility contract checked at gate CP1. The twin exposes itself to the loop through exactly three calls: `scan`, `truth_snapshot` (read-only, for evaluation), and `apply_plan` (which delivers a new dose and evolves the ground truth).

Synthetic acquisition and degradation. A scan is a biexponential IVIM forward model over nine b-values with Rician noise at nominal SNR 30, repeated to give a raw posterior spread. A vertical band clips the tumour where local SNR collapses to 7 — the *acquisition-degradation zone*. There the measured error bar on f genuinely inflates; this is the honest, *measured* signal the trust gate fires on (Sec. 4.6).

Dose/response. A voxel boosted above baseline loses perfusion, $f \leftarrow f\, e^{-\beta\, b_{\text{frac}}}$, where b_{frac} is the boost fraction; sparing or holding leaves f unchanged. This is the mechanism that lets the loop *converge*: treated tumour perfusion decays toward the spare threshold, after which the action gate stops boosting. It is a synthetic response model, not a radiobiological claim.

3.2. Posterior stage (Matrix's own estimator)

Each scan is fit with the textbook two-step *segmented* IVIM method: a high-b log-linear fit for D and the intercept-implied f, then a low-b residual fit for D^*. Fitting every noise realisation gives a raw per-voxel posterior centre μ and spread σ for (D, D^*, f). The estimator is deliberately the standard segmented method so that D^* (and, through the f–D^* coupling, f) is genuinely ill-conditioned in the high-D^* regime: nothing is hand-tuned to fake that (Sec. 4.6).

3.3. The four-stage loop

A shared per-iteration `LoopState` is threaded through four pure stages: (1) *scan* reads the truth snapshot and acquires a noisy scan; (2) *posterior* fits the raw (μ, σ) and applies the **ruler** to produce a calibrated σ_f with coverage/ECE; (3) *gates* apply the **trust gate** (which voxels are trustworthy) and the **action gate** (treat/spare/escalate, with action suppressed where untrustworthy); (4) *dose* applies the **dose engine**'s re-plan and delivers it, evolving the twin. The three consumed components are bundled in a single `Interfaces` object; swapping a real component for its placeholder is a one-line change *in the bundle*, never in a stage. This is the structural property Ferry later exploits (Sec. 5).

Trust gate. A voxel is flagged untrustworthy when its calibrated perfusion-fraction error bar is too wide ($\sigma_f > \sigma_{f,\max}$) or the fit lands in the high-D^* identifiability regime. The action gate is a Bayes-action-shaped interval rule on f — TREAT if the calibrated interval lies confidently above the treat threshold, SPARE if confidently below a spare threshold, ESCALATE otherwise — and the trust gate then *suppresses action* by forcing every untrustworthy voxel to ESCALATE regardless of the raw decision. The pre-suppression decision is recorded so the gate's effect can be measured.

3.4. Evaluation

Every load-bearing summary number is reported with a seeded bootstrap confidence interval; the trust gate's separation is summarised by the rank statistic AUROC of the calibrated σ_f against the (known) low-SNR label. All metrics operate only on the twin's known ground truth and the per-iteration state records; they make no clinical claim.

3.5. Interface design and the byte-identity anchor

The loop engine `run_iteration` references the twin only as a parameter and the three components only through their interfaces. This is what makes both the component-swap (placeholder → real Fashion/Minos/Forge) and the substrate-swap (synthetic twin → Ferry's grounded twin) possible without editing a single loop stage. We make the substrate-swap claim concrete and falsifiable: the loop engine's source file is *byte-for-byte identical* across the synthetic and grounded runs (git blob `4a34806ac4fa`), asserted by the Ferry drop-in gate (Sec. 5).

4. Results: the synthetic twin

All synthetic-twin results use a 12×12 twin, seed 20260622, 6 iterations, nominal SNR 30 (low-SNR zone 7), with 95% bootstrap CIs.

4.1. The loop closes

The four stages run every iteration, the twin evolves under the delivered plan, and a re-run from the same seed is bit-identical: the loop closes reproducibly and end-to-end. This is a property of the harness and twin and does not depend on any consumed component.

4.2. The calibrated error bar

The ruler[P] brings the perfusion-fraction error bar to near-nominal coverage on the twin: 95% empirical coverage 0.944 [0.903, 0.979] (seeded bootstrap CI over voxels) with expected calibration error $\mathrm{ECE}_f = 0.007$. This number rides on the Fashion placeholder and is marked [P]; the real Fashion ruler drops in behind the same interface.

4.3. The trust gate, and why $AUROC = 1.00$ is a property of the design

The trust gate flags the untrustworthy low-SNR zone with AUROC 1.00 of the calibrated σ_f against the known low-SNR label, firing at rate 1.00 inside the zone and 0.02 (CI hi 0.046) outside.

Scoping. An AUROC of 1.00 is **not** a performance claim. The synthetic twin *constructs* the untrustworthy zone (the low-SNR band) and the gate is keyed on the error bar that the same zone inflates; the two are separable *by construction.* The right reading is: "when an untrustworthy region is present and measurable, the gate keyed on the measured error bar identifies it," which is a sanity check on the harness, not evidence that any gate would separate trustworthy from untrustworthy voxels in real data. The value of grounding on real geometry (Sec. 5) is precisely that it stops this number from being read as more than it is.

4.4. Action suppression

Across the whole run, the trust gate holds the *action* rate on untrustworthy voxels at 0.000[P], versus 0.422 [0.362, 0.486][P] without the gate, while trustworthy voxels stay free to act (0.573 [0.536, 0.611][P]). The gate has a real, measured effect and does not over-suppress the trustworthy population.

4.5. Convergence

Trusted-tumour perfusion falls 0.176 [0.167, 0.185][P] over the loop (CI excludes 0) while the held untrusted tumour does not move (0.000): the honest cost of untrustworthiness is that those voxels are not treated. The treatment decision winds down — TREAT actions $21 \to 3 \to 0 \to 0 \to 0 \to 0$ — as mean true perfusion falls from 0.121 to 0.092 and dose de-escalation is bounded by the floor (40 Gy). Figure 1 shows the trust gate firing on the measured low-SNR zone and the closed-loop trajectory; Table 1 collects the load-bearing numbers.

4.6. An identifiability result: D^* cannot drive the gate; the gate is re-keyed on σ_f

The twin embeds a physically-motivated high-D^* sub-region, and one might expect the trust gate to flag it as untrustworthy. It cannot — and this is the point. Under the segmented fit, D^* is not identifiable: the low-b residual that carries D^* is dominated by noise, so the estimated D^* does not track the true high-D^* label. A gate keyed on

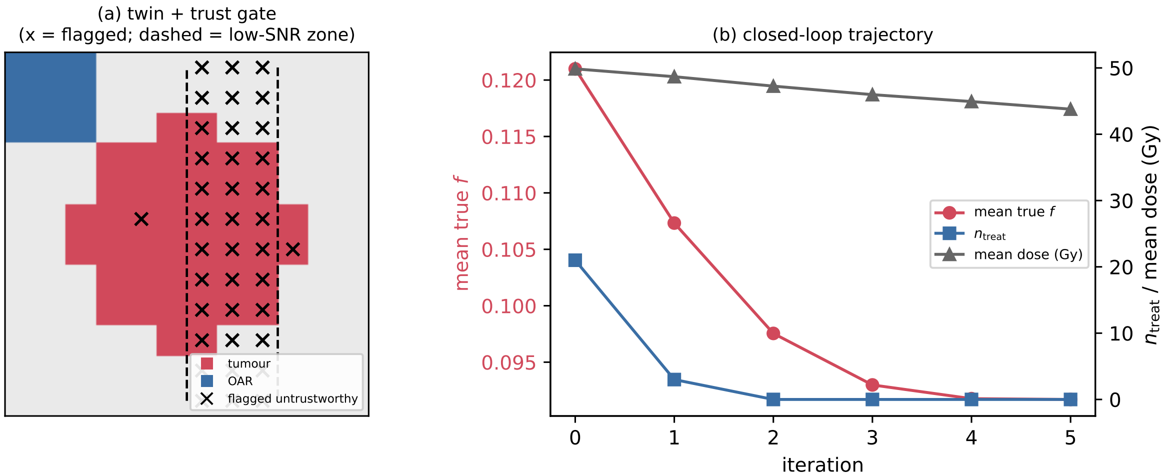

Figure 1. Synthetic twin (seed 20260622). **(a)** The trust gate (`x`) flags the measured low-SNR zone (dashed) on the tumour, which is why the AUROC of 1.00 is a property of the construction, not a performance claim (Sec. 4.3). **(b)** The closed loop converges: mean true f falls, TREAT actions wind down to zero, and dose de-escalates toward the floor. Generated by `paper/figures/make_figures.py` from the seeded run; numbers consuming Fashion/Minos are [P].

quantity	value [95% bootstrap CI]	rides on
trust-gate AUROC (σ_f vs low-SNR)	1.00 (by construction)	design
ruler coverage of f (95%)	0.944 [0.903, 0.979]	Fashion [P]
ruler ECE$_f$	0.007	Fashion [P]
action rate, untrustworthy, gated	0.000	Fashion+Minos [P]
action rate, untrustworthy, ungated	0.422 [0.362, 0.486]	Fashion+Minos [P]
action rate, trustworthy, gated	0.573 [0.536, 0.611]	Fashion+Minos [P]
trusted-tumour f drop	0.176 [0.167, 0.185]	Fashion+Minos [P]
untrusted-tumour f drop (held)	0.000	design

Table 1. Synthetic-twin load-bearing numbers (seed 20260622, 6 iterations). [P] marks numbers that consume the Fashion ruler and/or Minos gates (in review); the AUROC and the held-drop are properties of the synthetic design.

estimated D^* would therefore be keyed on noise. The working trust signal is instead the *measured* inflation of the calibrated perfusion-fraction error bar σ_f in the low-SNR zone — which is why the AUROC in Sec. 4.3 is computed on σ_f, not on D^*. We state this as an *identifiability* result scoped to the segmented-fit regime: high-D^* is the *failed* trust driver, and σ_f is the one that works. A different estimator (e.g. a full Bayesian fit) could change what is identifiable; the claim is about the segmented fit, not about IVIM in general.

5. Results: grounding on real anatomy and dose (Ferry)

To pre-empt the objection that the loop is only shown on a tidy synthetic twin, we ground it on **real** anatomy and a **real** delivered-dose grid from a public radiotherapy dataset — TCIA Pancreatic-CT-CBCT-SEG (DOI `10.7937/TCIA.ESHQ-4D90`, CC BY 4.0), patient `Pancreas-CT-CB_001`, an axial slice at $z = 316.91$ mm with real delivered dose in [13.72, 73.5] Gy. The grounded substrate ("Ferry") replaces the twin's synthetic anatomy and dose with the dataset's RTSTRUCT contours and RTDOSE grid; *perfusion/IVIM remains synthetic*, drawn with the identical seeded mechanism as the twin, so the *only* differences from the synthetic run are the two real fields.

Drop-in, proven byte-for-byte. Ferry is an interface-only swap: its grounded twin satisfies the same three-call contract the loop consumes, so it is handed to the *same* loop engine. We assert this structurally — the loop engine source is byte-for-byte identical to the synthetic run (git blob `4a34806ac4fa`) — so any behaviour change is attributable to real geometry and nothing else.

The loop closes on real geometry. On a 32×32 matched grid, the grounded loop closes: it runs end-to-end and reproducibly; the trust gate holds the untrustworthy *action* rate at 0.000[P] (vs 0.385 [0.360, 0.411][P] ungated); trusted-tumour perfusion falls 0.169 [0.166, 0.171][P] (CI excludes 0); and the treatment decision winds down ($231 \to 33 \to 4 \to 1 \to 0 \to 0$). Real anatomy is larger and irregular (477 vs 284 tumour voxels; 303/174 trusted/untrusted), and the loop's qualitative behaviour is preserved. The harness is not an artefact of the synthetic twin's tidy geometry.

5.1. The central honest negative: action-suppression is not outcome-protection

The grounded run surfaces a finding the synthetic twin cannot show, and it is the most important result in this paper (Fig. 2).

On the synthetic twin (and on real anatomy with a flat-baseline dose), "holding" an untrusted voxel leaves its perfusion untouched: the held drop is 0.000. On **real delivered dose**, the held untrusted tumour perfusion *still drops* — 0.148 [0.139, 0.156], CI excludes 0 — because the real prescription has *already* delivered dose to those voxels, which the twin's (synthetic) dose/response model then turns into a perfusion decline. The drop is synthetic perfusion responding to *real dose geometry*, not a measured biological effect; what matters is the loop logic it exposes. The trust gate suppresses *new actions*; it cannot undo dose already delivered.

The consequence is sharp: **action-suppression is not the same as outcome-protection**. A trust gate that withholds re-planning still leaves an untrusted voxel exposed to whatever dose the standing plan delivers. This is visible only once the dose geometry is real — on a flat synthetic baseline the held drop is identically zero, which is exactly why a synthetic-only study would have missed it. For an adaptive loop the

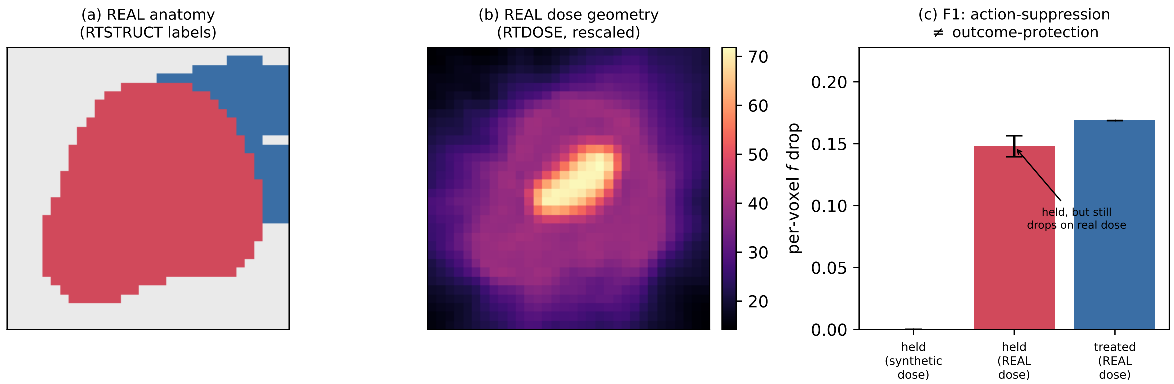

Figure 2. Grounding on real anatomy + dose (TCIA Pancreatic-CT-CBCT-SEG, patient `Pancreas-CT-CB_001`; CC BY 4.0). **(a)** Real RTSTRUCT anatomy (tumour, OAR) and **(b)** the real RTDOSE grid drive the loop in place of the synthetic twin's flat baseline. **(c)** The honest negative: a held untrusted voxel's f does not move under the synthetic flat dose (0.000) but *does* drop under *real* delivered dose (0.148 [0.139, 0.156], CI excludes 0), because the standing prescription already delivered dose there. The drop is synthetic perfusion responding to real dose geometry, not a biological measurement; [P] numbers consume Fashion/Minos.

metric	synthetic	grounded:anatomy	grounded:full
tumour voxels	284	477	477
trusted-tumour f drop	0.170	—	0.169 [0.166, 0.171]
untrusted (held) f drop	0.000	0.000	**0.148** [0.139, 0.156]
action rate untrust, gated	0.000	0.000	0.000
action rate untrust, ungated	0.423	—	0.385

Table 2. Ferry side-by-side on a matched 32×32 grid. The held untrusted f drop is 0.000 until *real dose geometry* enters (grounded:full), where it becomes 0.148 [0.139, 0.156] — the F1 honest negative. Numbers consuming Fashion/Minos are [P] (omitted here for the across-substrate comparison; full CIs in `results/RESULTS_FERRY_CP2.md`).

implication is that trust-gating the *decision* is necessary but not sufficient; protecting an outcome additionally requires control over the standing prescription, which is the dose engine's responsibility (Forge, future work). Table 2 places the finding in its side-by-side context. Crucially, this negative does *not* depend on the Fashion or Minos placeholders being correct: it follows from real delivered dose plus the gate's structural property of acting only on *new* actions, so it survives a faithful drop-in of the real components.

What a reviewer can and cannot conclude. A reviewer *can* conclude that Matrix's closed loop closes on real geometry. A reviewer *cannot* conclude anything about real perfusion or IVIM: the (D, D^*, f) ground truth, the scan, and the SNR map are synthetic. Ferry closes the *geometry* gap; the residual *diffusion* gap — real multi-b acquisition on the same anatomy — is closed only by a scanner, which is Keystone's real-time/offline mode and explicitly out of scope here.

6. Discussion

The honest negative is the contribution. It would be easy to report only the positives: the loop closes, the gate separates perfectly, perfusion falls under treatment. The most useful result is the negative one. F1 (Sec. 5.1) says that a trust gate, however well it suppresses action, does not by itself protect an untrusted voxel from a standing dose — and it says so with a CI that excludes zero, on real delivered dose, against a synthetic baseline that is identically zero. Surfacing that distinction early, on a synthetic-plus-grounded harness, is cheaper and safer than discovering it in a clinical loop.

Identifiability, not impossibility. The D^* result (Sec. 4.6) is scoped: under the segmented fit, D^* is not a valid trust driver, so the gate is re-keyed on the measured σ_f. This is an identifiability statement about a specific estimator, not a claim that D^* is unknowable in principle. A richer posterior could change what is identifiable; the harness is built so that swapping the estimator is a localized change.

What is solid and what is provisional. "The loop closes" — the harness, the twin, the segmented posterior, and the Ferry geometry result — is solid for what it claims. The calibration, trust, and convergence *numbers* consume the Fashion ruler and Minos gates and are provisional ([P]) until those publish; their interfaces are fixed, so a faithful drop-in re-runs the same gates. The F1 negative is robust to that drop-in by construction. **Forge** (the dose engine) is not built; its placeholder affects only the physical realism of the re-plan (e.g. a NOT-Forge warrant artefact on a non-uniform real prescription), never whether the loop closes or how the gates behave. We present Forge as drop-in future work, not a dependency met, and it is not a release condition for this manuscript.

Relation to Keystone. Matrix is the no-scanner run-mode of a closed-loop capstone. Its value is that the control logic and the trust-gating discipline can be validated — and their honest limits surfaced — before any scanner, dose engine, or clinical data is in hand.

7. Scope ceiling and limitations

We restate the boundary because it is load-bearing. (1) *No clinical claim.* Every result means "the loop closes and behaves sensibly," on a synthetic twin or on real geometry with synthetic perfusion. (2) *Synthetic perfusion everywhere.* Even under Ferry, (D, D^*, f), the scan, and the SNR map are synthetic; there is no scanner and no real diffusion data. (3) $AUROC = 1.00$ *is by construction* (Sec. 4.3), not a performance claim. (4) *Provisional components.* Calibration/trust/convergence numbers ride on Fashion and Minos (in review) and are marked [P]; the dose engine (Forge) is not built.

(5) *Identifiability is scoped* to the segmented fit (Sec. 4.6). The single residual gap that turns any perfusion reading into a clinical statement — real IVIM acquisition — is left to a future scanner study.

8. Conclusion

Matrix closes a four-stage adaptive-radiotherapy control loop on a synthetic digital twin, withholds action where the calibrated perfusion-fraction error bar is untrustworthy, and converges — all reproducibly and with confidence intervals against known ground truth. It grounds the same loop, byte-for-byte unchanged, on real anatomy and real dose geometry, and in doing so surfaces an honest negative that a synthetic-only study would miss: action-suppression is not outcome-protection on a real prescription. As a methods demonstration it validates a harness and a trust-gating discipline — and their limits — not any clinical effect.

Reproducibility and data availability

Every number in this manuscript is regenerated from seeded results by `paper/consistency.py` (which writes `numbers.tex`) and is traceable to a gate printout: `results/RESULTS_CP2.json` and `RESULTS_CP4.json` (synthetic twin, regenerated offline by `reproduce.sh`) and `RESULTS_FERRY_CP2.json` (grounded run, regenerated from the public TCIA dataset and committed). The loop engine's byte-identity across substrates (git blob `4a34806ac4fa`) is asserted by the Ferry drop-in gate. The real anatomy/dose are from TCIA Pancreatic-CT-CBCT-SEG (DOI `10.7937/TCIA.ESHQ-4D90`, CC BY 4.0, attribution only); no patient data or image blobs are committed — the loader fetches and caches them by script. No private or clinical data is used anywhere.

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[2] A. Karlin, *Minos: the decision value of a trusted gate (VoTG) for quantitative-MRI treat/spare/escalate*. In review, 2026.
[3] A. Karlin, *Forge: a Monte-Carlo dose engine for MR-guided adaptive RT* (feasibility benchmark; re-plan engine deferred to 2027).
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